(d) Die draaimoment τ in die vorige vraag veroorsaak dat die liggaam 'n hoekversnelling α ondergaan. Met $\theta = 90°$, toon aan dat $\tau = I\alpha$.
In the previous question the torque τ results in an angular acceleration α. With $\theta = 90°$, show that $\tau = I\alpha$.

$\tau = rF \sin 90$

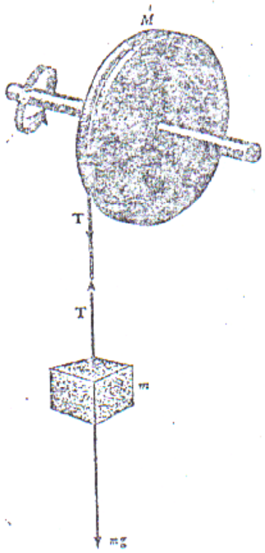
$\tau = r \times F = rma$

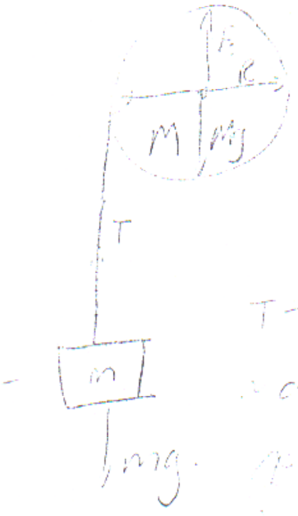
$= mr^2 \alpha$

$= I\alpha$

(4)

(e) 'n Wiel met straal R, massa M en traagheidsmoment I roteer om 'n as. 'n Massa m is aan die wiel verbind, soos in die skets.
A wheel with radius R, mass M and moment of inertia I rotates about an axis. A mass m is connected to the wheel, as shown in the figure.





$T - mg =$

$ma = mg$

$a = r\alpha$

$\tau = TR$

(i) Herlei 'n uitdrukking vir die hoekversnelling α in terme van onder andere die spankrag T in die tou.
Derive an equation for the angular acceleration α in terms of for instance the tension T in the rope.

$\alpha = \frac{d\omega}{dt} = \frac{d}{dt}\frac{v}{r} = r\frac{dv}{dt} = r$ $\qquad \alpha = \frac{TR}{I}$

(2)

Vraag / *Question* 4(e)(ii)(Vervolg)(*Continue*) / .....